Problem 4: Exam Score Band Counter

TASK

A university stores the marks of all students in an exam as an array sorted in ascending order; marks can repeat. Faculty members frequently ask how many students scored within a band [low, high], inclusive of both ends.

Implement a function countInBand(scores, low, high) that returns the number of scores s such that low <= s <= high. The array is large and queried very often, so your solution must not scan the scores one by one.

CONSTRAINTS

- 1 <= length of scores <= 10^6
- scores is sorted in ascending order
- 0 <= low <= high <= 100

EXAMPLES

Sample 1

```
scores = [35, 42, 42, 50, 58, 58, 58, 63, 71, 88], low = 42, high = 58

Expected Output: 6
```

Sample 2

```
scores = [35, 42, 42, 50, 58, 58, 58, 63, 71, 88], low = 90, high = 100

Expected Output: 0
```

COMPLEXITYREQUIREMENTS

State the time and space complexity of a linear scan and of your final solution. Explain how duplicates at the edges of the band affect the way your search must be written.

HINT

A standard binary search stops at any matching element. Modify it to find the first position with a score >= low and the first position with a score > high; the answer is the difference.